

# torch.sparse.sampled\_addmm#

[https://pytorch.org/docs/stable/generated/torch.sparse.sampled\\_addmm.html](https://pytorch.org/docs/stable/generated/torch.sparse.sampled_addmm.html)

## Description:

Performs a matrix multiplication of the dense matrices `mat1` and `mat2` at the locations specified by the sparsity pattern of `input`. The matrix `input` is added to the final result. Mathematically this performs the following operation: where  $\text{spy}(\text{input})$  is the sparsity pattern matrix of `input`,  $\alpha$  and  $\beta$  are the scaling factors.  $\text{spy}(\text{input})$  has value 1 at the positions where `input` has non-zero values, and 0 elsewhere. `input` must be a sparse CSR tensor. `mat1` and `mat2` must be dense tensors.

## Function Signature

```
torch.sparse.sampled_addmm(input, mat1, mat2, *, beta=1.,  
alpha=1., out=None) → Tensor#
```

Parameters

Keyword Arguments

## Parameters

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### Keyword

beta (Number, optional) – multiplier for input ( $\beta$ ) alpha (Number, optional) – multiplier for  $\text{mat1} @ \text{mat2}$   $\text{mat1} @ \text{mat2}$  ( $\alpha$ ) out (Tensor, optional) – output tensor. Ignored if None. Default: None.

## Code Examples

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```
>>> input = torch.eye(3, device='cuda').to_sparse_csr()
>>> mat1 = torch.randn(3, 5, device='cuda')
>>> mat2 = torch.randn(5, 3, device='cuda')
>>> torch.sparse.sampled_addmm(input, mat1, mat2)
tensor(crow_indices=tensor([0, 1, 2, 3]),
      col_indices=tensor([0, 1, 2]),
      values=tensor([ 0.2847, -0.7805, -0.1900]), device='cuda:0',
      size=(3, 3), nnz=3, layout=torch.sparse_csr)
>>> torch.sparse.sampled_addmm(input, mat1, mat2).to_dense()
tensor([[ 0.2847,  0.0000,  0.0000],
        [ 0.0000, -0.7805,  0.0000],
        [ 0.0000,  0.0000, -0.1900]], device='cuda:0')
>>> torch.sparse.sampled_addmm(input, mat1, mat2, beta=0.5,
alpha=0.5)
tensor(crow_indices=tensor([0, 1, 2, 3]),
      col_indices=tensor([0, 1, 2]),
      values=tensor([ 0.1423, -0.3903, -0.0950]), device='cuda:0',
      size=(3, 3), nnz=3, layout=torch.sparse_csr)
```

## Notes & Warnings

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### NOTE

Note input must be a sparse CSR tensor. mat1 and mat2 must be dense tensors.

# Detailed Documentation

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## Documentation

Performs a matrix multiplication of the dense matrices `mat1` and `mat2` at the locations specified by the sparsity pattern of input. The matrix input is added to the final result.

Mathematically this performs the following operation:

where  $\text{spy}(\text{input})$  is the sparsity pattern matrix of input,  $\alpha$  and  $\beta$  are the scaling factors.  $\text{spy}(\text{input})$  has value 1 at the positions where input has non-zero values, and 0 elsewhere.

input must be a sparse CSR tensor. `mat1` and `mat2` must be dense tensors.

input (Tensor) – a sparse CSR matrix of shape (m, n) to be added and used to compute the sampled matrix multiplication

mat1 (Tensor) – a dense matrix of shape (m, k) to be multiplied

mat2 (Tensor) – a dense matrix of shape (k, n) to be multiplied

beta (Number, optional) – multiplier for input ( $\beta$ )

alpha (Number, optional) – multiplier for  $\text{mat1} @ \text{mat2}$  ( $\alpha$ )

out (Tensor, optional) – output tensor. Ignored if None. Default: None.